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The Traveling Salesman Problem for lines in the plane

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Abstract

We present an algorithm that computes the shortest route that intersects n lines in the plane in time polynomial in n ; the time complexity is dominated by the time it takes to compute a shortest watchman route in a simple polygon of size n . We also present a simple $O(n)$ time algorithm for computing a route that intersects the lines but which is guaranteed to have a length at most $\sqrt{2}$ times longer than the shortest route. © 2002 Elsevier Science B.V. All rights reserved.

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1. Introduction

Path planning is one of the central problem areas in computational geometry and robotics. One particular kind of path planning problems which has been studied involves ideal (point-sized and arbitrary turning) robots required to touch or visit given sets of objects. In their most general form, we call the solutions to these problems *visiting paths*. Maybe the most classical example of a shortest visiting path is the *route* — i.e., the closed path — asked for in the *Traveling Salesman Problem* (TSP) [6]. Although this famous problem, in which the solution is the shortest visiting route for a given set of points in the plane, is NP-complete [10], recent studies have revealed that many variations of the problem can be solved in polynomial time [5]. Moreover, for some of the hard variants, for instance the *TSP with Neighborhoods* [9] which is NP-hard, there are efficient algorithms that compute provably good approximations to the exact solutions.

In this paper we introduce and study the *Traveling Salesman Problem for lines*, a variation of the TSP in which the shortest visiting route for a set of n arbitrarily slopped lines in the plane is asked for. In Fig. 1, an example of a shortest visiting route for 8 lines is shown. In Section 2 we show how an exact solution can be computed in polynomial time by reducing the problem to the problem of computing a shortest watchman route in a simple polygon [4]. The reduction is based on the construction of a special simple polygon of size $O(n)$ defined by the lines and built in $O(n \log n)$ time. In Section 3 we then present a simple and robust $O(n)$ time algorithm for computing a route that intersects the lines but which is guaranteed to have a length at most $\sqrt{2}$ times longer than the shortest route. Fig. 2 shows the smallest perimeter axis-parallel rectangle that intersects the lines in Fig. 1.

2. Computing an exact solution

In this first part of the paper, we study the following problem: “Given a (*finite*) set of lines in the plane,

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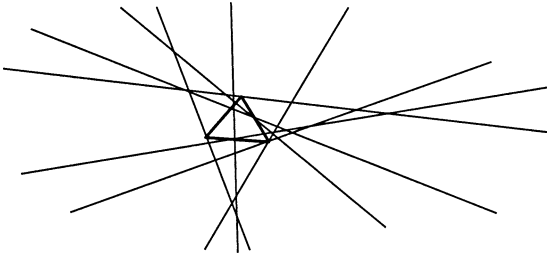


Fig. 1. A shortest visiting route.

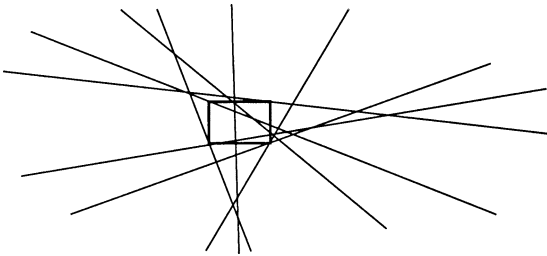


Fig. 2. The smallest perimeter axis-parallel rectangle that intersects all 8 lines in Fig. 1.

compute the shortest route that visits them”. Throughout this paper, the given lines are referred to as *target lines*. Furthermore, $\mathcal{L}$ denotes the set of target lines while n denotes the size of $\mathcal{L}$. The word “visits” is synonymous with “intersects” but its use is reserved for intersections between paths and target lines.

To solve the problem given above we make use of an observation that relates it to the problem of computing a shortest watchman route in a simple polygon. In the latter problem, a closed path in a simple polygon such that each point in the polygon is visible from some point on the path is asked for. Here, two points in a simple polygon are *visible* to each other if the line segment between the points does not intersect the exterior of the polygon. This problem can be solved in linear time [7] if the polygon is *star-shaped*, that is if it contains a point from which all other points in the polygon are visible. Then this point is a shortest (degenerate) watchman route. In polygons which are not star-shaped a shortest watchman route can be computed in $O(n^5)$ time.²

² Recently, a flaw in an algorithm for computing a *fixed* shortest watchman route was corrected [12]. Since this algorithm is used as a subroutine in the algorithm by Carlsson et al. [3], the complexity of the latter algorithm increased from the originally stated $O(n^4)$ by a

It is known that each non-star-shaped polygon contains a finite set of line segments that it is necessary and sufficient for a route to visit in order to be a watchman route [3]. The observation we use is that the shortest watchman route is a shortest visiting route for these line segments in the polygon. Suppose that the shortest watchman route does not touch the boundary of the polygon. Then it is in fact a shortest visiting route not only for the line segments themselves (disregarding the polygon) but also for the lines in the plane that contain the line segments.

In the remainder of this section, we define a simple polygon P in terms of the target lines in $\mathcal{L}$ such that a shortest watchman route in P is a shortest visiting route for the target lines. We then show how P , and consequently a shortest visiting route for the target lines, can be computed in time polynomial in n .

2.1. Shortest visiting routes for target lines

We will use the symbol V_{opt} to denote an arbitrary route among the potentially infinitely many shortest visiting routes for the set $\mathcal{L}$. The fact that V_{opt} is the shortest possible route has several implications. As for its general shape, the parts between consecutive points of visit (*vertices*) on target lines are line segments (*edges*), since they are shortest paths in the plane. Moreover, since a non-convex route in the plane is strictly longer than the boundary of its convex hull, and the convex hull of a route intersects any target line the route intersects, we conclude that V_{opt} is convex. This implies that V_{opt} visits each target line in one of three ways. It *crosses over*, or makes a *crossing contact* with, a target line if it not only shares at least one point with the line but also contains points in each of the two half-planes separated by the line (Fig. 3(a)). If V_{opt} does not cross over a target line, there is, since V_{opt} is convex, exactly one connected subset of the target line where V_{opt} visits it. If the subset is a single point, this point is a *point of reflection* and the contact is referred to as a *reflection* (Fig. 3(a)). If it is a line segment, we have a *tangential contact* which can be thought of as a degenerate reflection (Fig. 3(b)). The convexity of V_{opt} also implies that V_{opt} reflects at most once on each target line. Such a reflection

factor $O(n^2)$. This result was very recently improved to $O(n^5)$ time by Tan [11].

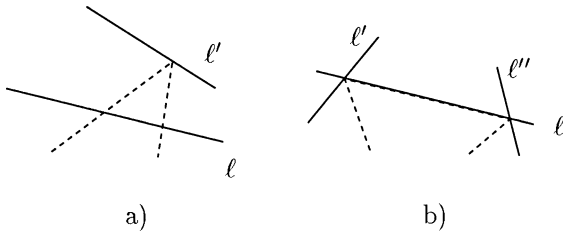


Fig. 3. (a) A crossing contact with ℓ and a reflection on ℓ' . (b) A tangential contact with ℓ .

occurs at a *crossing*, i.e., a point where two target lines intersect, or in *fragment*, which is a maximal subset of a target line that does not contain a crossing. Let the *angle of reflection* be the angle formed at a point of reflection by either the incoming or the outgoing edge of V_{opt} and the target line. Then, for reflections in fragments, the local optimality of V_{opt} is given by *Heron's reflection principle*:

Lemma 2.1 (The reflection principle). *The angles of reflection are equal at a point of reflection that lies in a fragment.*

A reflection where the two angles of reflection are equal is *perfect*. Since V_{opt} is the shortest possible route, we have:

Lemma 2.2. *Let v_i denote a vertex of V_{opt} and assume without loss of generality that V_{opt} is located so that it makes a perfect reflection on the horizontal line through v_i . Then v_i lies either in a horizontal target line ℓ (on which V_{opt} makes a perfect reflection) or at a crossing between two target lines with negative and positive slope, respectively.*

Proof. The lemma follows from the fact that if neither ℓ nor the two target lines with negative and positive slope, respectively, are among the target lines on which V_{opt} reflects at v_i , V_{opt} can be shortened by moving v_i along a target line. $\square$

To bound the region in the plane where V_{opt} lies in terms of the target lines in $\mathcal{L}$, consider the convex hull C (see Fig. 4) of all crossings.³ Since C is convex,

³ If there are no crossings, V_{opt} is merely a line segment perpendicular to the target lines.

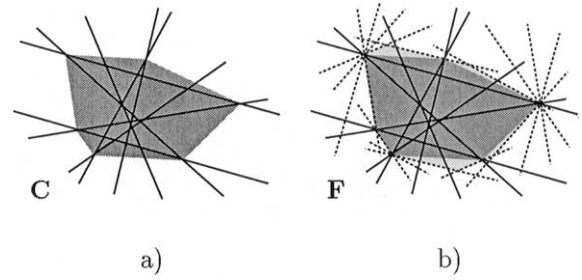


Fig. 4. (a) The convex hull C . (b) The polygon F (shaded and containing the even darker shaded hull C).

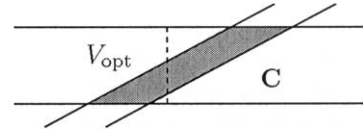


Fig. 5. V_{opt} is not always contained in C .

each of its tangents partitions the plane into two half-planes of which the *active half-plane* totally contains C . For sets $\mathcal{L}$ containing no parallel target lines, it can be shown that V_{opt} lies in C . However, in the presence of parallel target lines, the convex hull C does not always contain V_{opt} (Fig. 5). Consider the intersection F of the active half-planes of all tangents of C that are normals to target lines. For F we have:

Lemma 2.3. *V_{opt} is contained in F .*

Proof. Assume, in order to get a contradiction, that V_{opt} is not totally contained in the active half-plane of the tangent t of C that is a normal to a target line ℓ . Let t' denote the normal to ℓ that is a tangent to V_{opt} and does not lie in the active half-plane of t . Being a tangent, t' intersects V_{opt} at a vertex v_i that, since V_{opt} is the shortest possible, lies in a target line ℓ_i . To simplify the presentation, we assume without loss of generality that ℓ is vertical, that v_i lies to the right of ℓ , and that V_{opt} lies below t' (Fig. 6).

The vertex v_i belongs to no other target line than ℓ_i since it lies in the exterior of C where there are no crossings. Lemma 2.1 then implies that the reflection at v_i must be perfect, which in turn implies that ℓ_i is not parallel to ℓ since v_i would then not be a topmost vertex of V_{opt} . Rather, ℓ_i has a positive slope less than infinity since it must intersect ℓ in C and below t .

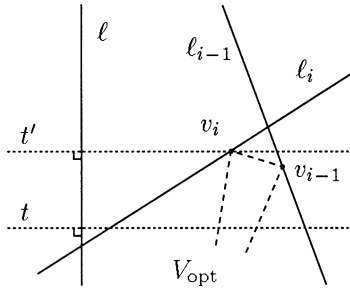


Fig. 6. Illustration used in Lemma 2.3.

Moreover, the angle of reflection at v_i can not be exactly $\pi/2$ since V_{opt} would in that case not intersect ℓ . We conclude that the angle is less than $\pi/2$, from which follows that V_{opt} has at least three vertices. Then, by Lemma 2.2, there is a target line through the vertex of V_{opt} that occurs immediately before v_i in the counter-clockwise direction and intersects ℓ_i above t' . However, then this crossing lies in the exterior of C in contradiction to the fact the C contains all crossings. From this the lemma follows. $\square$

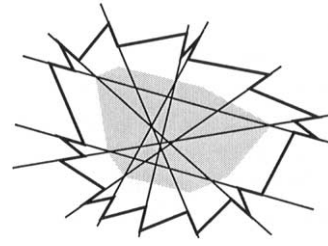
2.2. The polygon P

P can be described as a convex polygon that is big enough to contain F and to whose boundary sawtooth-like notches have been added wherever target lines intersect. An example is shown in Fig. 7 and the technical details are as follows.

Let $\{f_0, f_1, \dots, f_{2n-1}\}$ be the set of end points of the intersections between F and the target lines, where the indices denote their counter-clockwise ordering along the boundary of F , and let $\delta > 0$ be a constant smaller than the length of the shortest of these intersections. Also, let ω_i denote the index of the target line that contains the intersection for which f_i is an end point. Then, the polygon P is the region bounded by the concatenation $\bigcup_{i=0}^{4n-1} [p_i, p_{(i+1)}]$ of line segments where $p_0 = p_{4n}$ and p_i is defined as follows for $0 \leq i \leq 4n - 1$:

- If i is even, p_i is the point on $\ell_{\omega_{[i/2]}}$ at distance $\delta/2$ from $f_{[i/2]}$ in the exterior of F ;
- if i is odd, p_i is instead the point on $\ell_{\omega_{[i/2]}}$ at distance δ from $f_{[i/2]}$ in the exterior of F .

Note that there are twice as many vertices in the boundary of P as there are end points f_i , and that the point at distance δ from $f_{[i/2]}$ in the exterior of F and

Fig. 7. The polygons P and F (shaded).

on $\ell_{\omega_{[i/2]}}$ is not necessarily unique if δ is larger than the length of the shortest line segment.

The boundary of P zig-zags around the interior of P . Every other vertex is reflex, i.e., the angle formed by the edges joined at the vertex is greater than π in P , while the others are convex (i.e., not reflex). For any even i , we call the sub-chain $[v_i, v_{i+1}] \cup [v_{i+1}, v_{i+2}]$ in the boundary of P a sawtooth. We have the following relationship:

Lemma 2.4. *A shortest watchman route in P is a shortest visiting route for $\mathcal{L}$ in the plane.*

Proof. Let C_i denote the chord in P that is formed as the intersection between P and ℓ_i . We first prove that it is necessary and sufficient for a route to visit all chords C_i to be a watchman route.

Since any chord C_i partitions P into two subpolygons, each of which is not totally visible from the interior of the other, it is necessary for a route to intersect all chords C_i in order to be a watchman route. Next, let W denote a route that intersects the two chords C_i and C_j that extend into P from the reflex vertices that bounds a sawtooth. We then claim that the sawtooth is visible from W . Assume that this is not the case, i.e., that there is a point p in the sawtooth that is not visible from W . Should C_i and C_j be parallel, the chord through p parallel to C_i would separate C_i and C_j from each other and be intersected by W , which would imply that p is visible from W in contradiction to our assumption. If, on the other hand, C_i and C_j intersect in P , W has to intersect the chord from p through the point of intersection, so p would be visible also in this case. From this our claim follows. Moreover, if the entire boundary of a simple polygon is visible from a route, the route is a watchman route [3]. Hence, it is

sufficient for a route to intersect all chords C_i to be a watchman route.

The necessity and sufficiency requirements imply that a watchman route in P is a visiting route for $\mathcal{L}$ in the plane and vice versa as long as the visiting route is contained in the interior of P . Hence, since by Lemma 2.3 all shortest visiting routes for $\mathcal{L}$ are contained in F , which is a subset of P , any shortest watchman route in P is a shortest visiting route for $\mathcal{L}$. $\square$

2.3. Computing V_{opt}

The algorithm that computes V_{opt} takes as input the set $\mathcal{L}$ and computes P as a first step. The convex hull C can be computed in $O(n \log n)$ time [1]. This involves sorting the target lines by slope as the primary key and vertical position as the second key, which causes parallel target lines to end up next to each other in the resulting ordering.

Since the target lines diverge in the exterior of C , they intersect the boundary of C in slope order. Given a target line, the tangent of C that bounds one of the active half-planes of the target line can be computed in $O(\log n)$ time by binary search along the boundary of C . This tangent passes through a vertex of C . Moreover, since C is convex, the tangent of C that is a normal to the next target line in the sorted order also passes through this vertex or some vertex further away along the boundary of C . Hence, once the first tangent has been computed, the vertices where the remaining tangents intersect the boundary of C follow after each other in sorted slope order along the boundary. Therefore, only a single scan in one direction around the boundary of C is needed in order to compute them all. It follows that the intersection F of their active half-planes can therefore be computed in $O(n)$ time [2]. Note that the order in which the target lines intersect the boundary of C and F is the same since both polygons are convex, $C \subset F$, and C contains all crossings. Hence, the vertices p_i defining the polygon P can be computed in $O(n)$ time as well.

From the description given above, it follows that P can be constructed in $O(n \log n)$ time and that P has $O(n)$ vertices. By Lemma 2.4, a shortest watchman route in P is a shortest visiting route for $\mathcal{L}$. Then, since a shortest watchman route in a simple polygon can be computed in $O(n^5)$ time [11], we have:

Theorem 1. *A shortest visiting route for n lines in the plane can be computed in $O(n^5)$ time.*

3. A linear time approximation

In this section we give a linear-time algorithm, based on linear programming, that computes the *shortest perimeter axis-parallel rectangle* R_{opt} that intersects all target lines in $\mathcal{L}$.

Lemma 3.1. *R_{opt} is a visiting route for $\mathcal{L}$ at most $\sqrt{2}$ times as long as V_{opt} .*

Proof. Let the *bounding box* of a route be the axis-parallel rectangle with shortest perimeter that contains the route. First, we claim that the bounding box of the (convex) route V_{opt} is at most $\sqrt{2}$ times longer than the route itself. Because of the minimality of bounding boxes, each of the four edges of the bounding box of V_{opt} touches V_{opt} in at least one point. Consider the shortest route that visits these four points of intersection. This route visits the points in order along the boundary of the bounding box, since a shortest route does not cross over itself, and is therefore a convex quadrilateral. Then, the boundary of the bounding box is at most $\sqrt{2}$ times longer than the shortest route since the sum of the lengths of the catheti is at most $\sqrt{2}$ times the length of the hypotenuse in a right-angled triangle. Hence, since V_{opt} is at least as long as the shortest route, the claim follows. Second, the minimality of R_{opt} implies that the length of the boundary of R_{opt} is at most as large as the length of the boundary of the bounding box of V_{opt} . Then the lemma follows since R_{opt} intersects all target lines. $\square$

In the linear program, we use the length of the perimeter of R_{opt} as the object function. In other words, we seek the coordinates x^- , x^+ , y^- , and y^+ of the vertices (x^-, y^-) , (x^-, y^+) , (x^+, y^-) and (x^+, y^+) of R_{opt} , where $x^- < x^+$, and $y^- < y^+$, which minimizes $2(x^+ - x^-) + 2(y^+ - y^-)$. Should the input contain vertical target lines, we eliminate these by first finding the smallest slope α_1 and the second smallest slope α_2 among the slopes of the target lines. Then, since any slope α between α_1 and α_2 is different from the slope of any target line,

rotating the axis so that α becomes the vertical slope ensures that no target line is vertical. Let $\mathcal{L}^-$ and $\mathcal{L}^+$ be the set of target lines with negative and positive slope, respectively. Then, since each edge of R_{opt} is parallel to one of the axes, we have the following lemma:

Lemma 3.2. *The target lines in $\mathcal{L}^-$ and $\mathcal{L}^+$ intersect the diagonals $[(x^-, y^-), (x^+, y^+)]$ and $[(x^-, y^+), (x^+, y^-)]$ of R_{opt} , respectively.*

Lemma 3.2 gives us the linear constraints for our program. Altogether, the problem of computing R_{opt} can now be formulated as follows: “Minimize $2(x^+ - x^-) + 2(y^+ - y^-)$ subject to: $by^+ + ax^- \geq -c$ and $-by^- - ax^+ \geq -c$ for all $ax + by + c = 0 \in \mathcal{L}^+$, $-by^- - ax^+ \geq c$ and $by^+ + ax^- \geq c$ for all $ax + by + c = 0 \in \mathcal{L}^-$, $x^+ - x^- \geq 0$, and $y^+ - y^- \geq 0$ ”. Since the program is formulated using $2n + 2$ constraints and 4 variables x^- , x^+ , y^- , and y^+ , and the solution to a linear program in a fixed number of variables can be found in time linear in the number of constraints [8], by Lemma 3.1, we get:

Theorem 2. *A visiting route for n lines in the plane, such that the length of the route is at most $\sqrt{2}$ times longer than a shortest visiting route for the lines, can be computed in $O(n)$ time.*

4. Conclusions and discussion

It is unclear whether or not the problem of finding a shortest route that intersects n given lines in the plane can be solved in less than $O(n^5)$ time. Although a lower bound for this problem is also a lower bound for the problem of computing a shortest watchman route, no explicit (non-trivial) lower bounds are known for either problem.

If the lines are replaced with line segments the problem turns into the TSP with Neighborhoods which is NP-hard. Should the lines instead be replaced with rays we get a variant of the problem that is open and lies somewhere in between the problems concerning lines and line segments. Note that in this case the reduction to the problem of computing a shortest watchman route used in Section 2 no longer works since rays that have their starting points in F do not

yield useful cuts for defining P . However, a $\sqrt{2}$ -approximation can still be obtained in $O(n)$ time by using the linear program presented in Section 3 but with a linear number of additional constraints to ensure that the starting points of the rays lie in R_{opt} .

Finally, we point out that the shortest visiting (non-closed) path for n lines can also be approximated by our algorithms. The route we get if we walk twice along a shortest visiting path for $\mathcal{L}$ (back and forth) is a visiting route for $\mathcal{L}$ twice as long as the path. Since this visiting route is not shorter than the shortest visiting route, it follows that a shortest visiting route for $\mathcal{L}$ is at most twice as long as a shortest visiting path for $\mathcal{L}$. Hence, by Theorem 2, we conclude that a visiting path for n lines in the plane, such that the length of the path is at most $2\sqrt{2}$ times longer than a shortest visiting path for the lines, can be computed in $O(n)$ time.

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